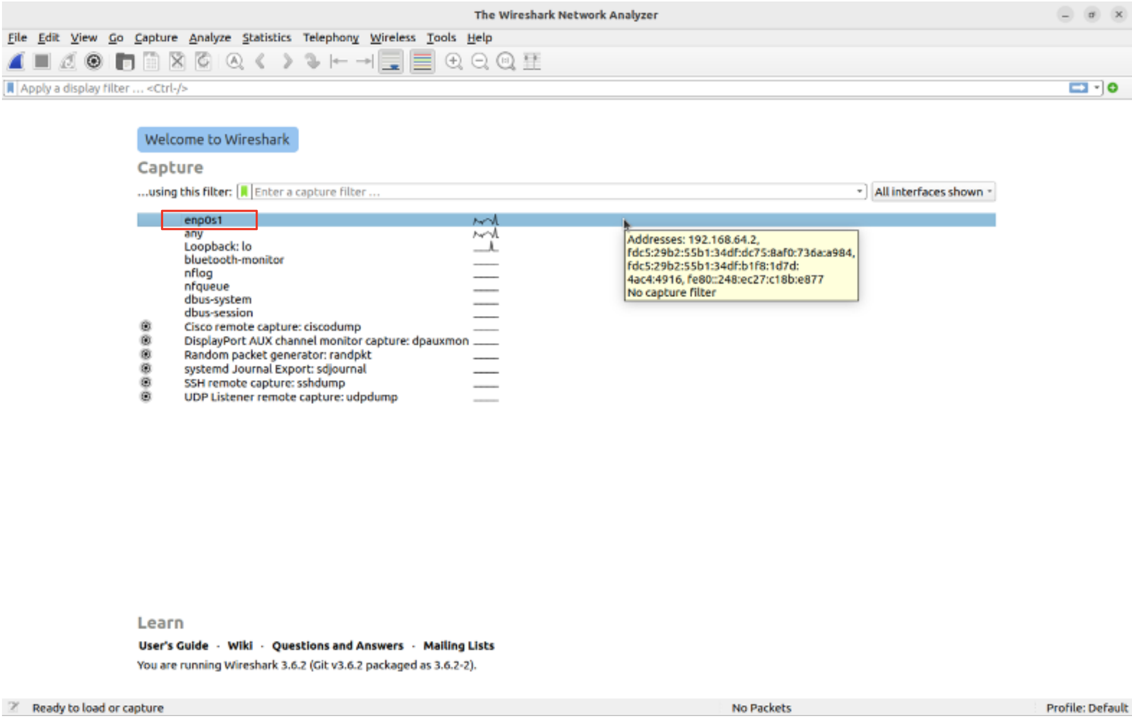
Yuval baror 214329633
Joshua gordon332307073

Question number 1:

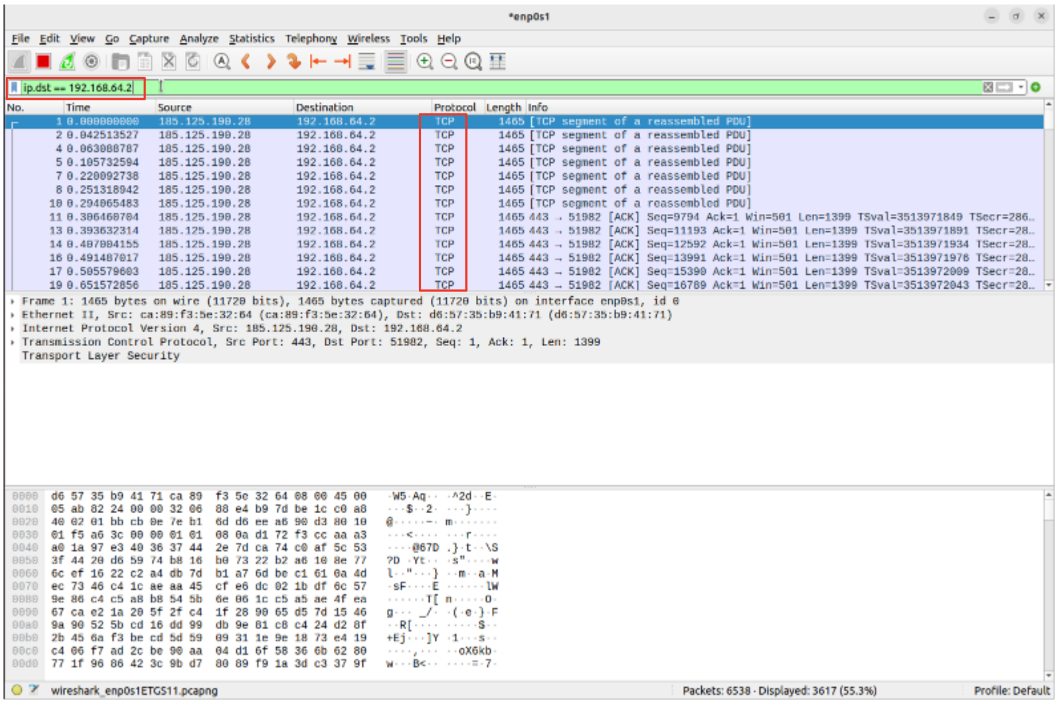


As one can see we have here opened wireshark as required, and we have our pointer on the interface that's connected to the external internet.

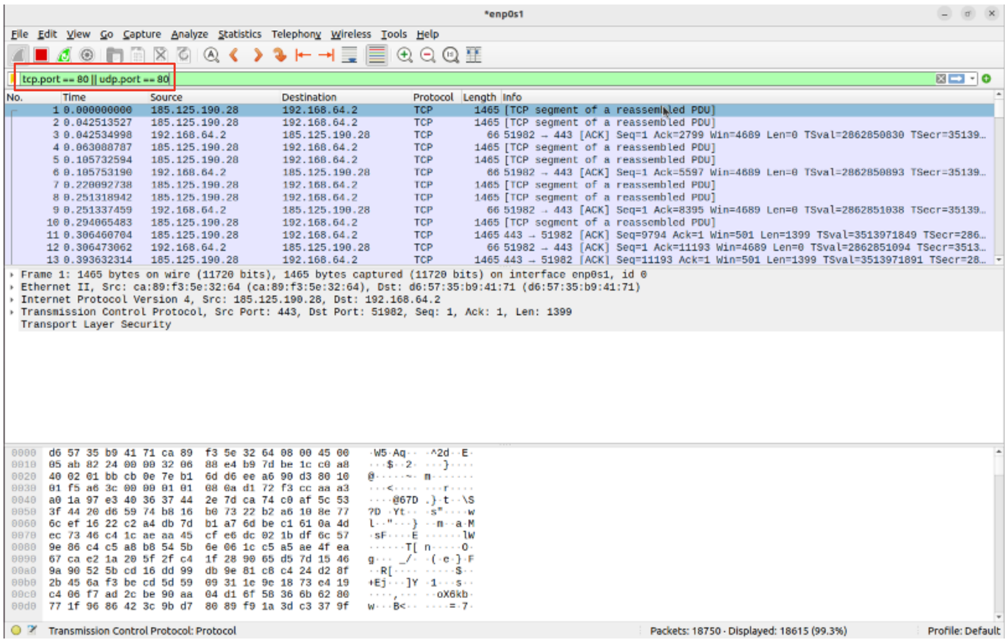
Question number 2:

A:

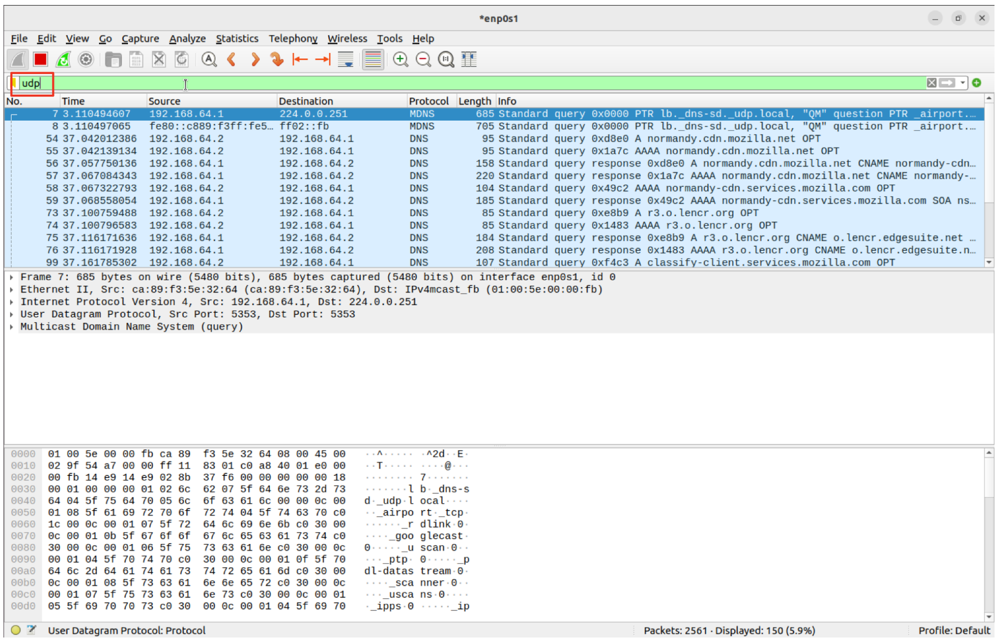
we shall filter the packets based on the ip destination based on the address 192.168.64.2 (Our chosen ip destination) with the correct search term in the command line: ip.dst == 192.168.64.2



B:
filtering the packets by source port 80. one can clearly see in the screenshot the protocol that is being used is tcp.

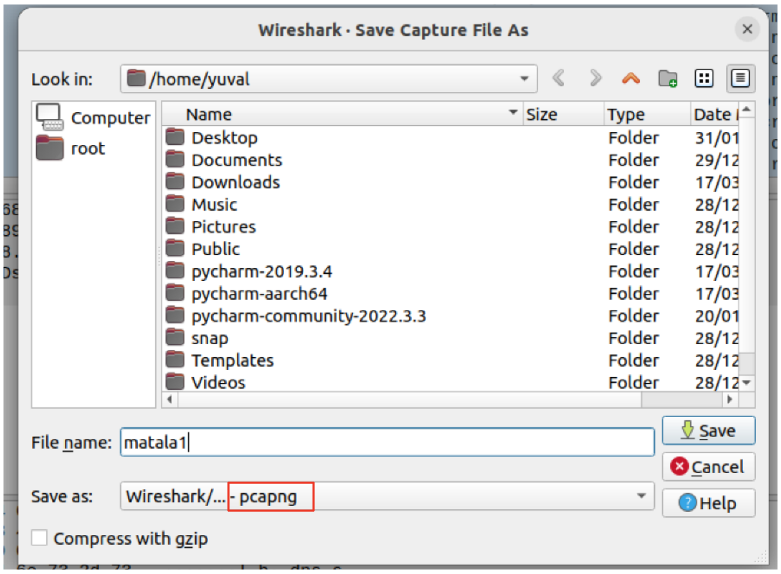


C:
Filtering for UDP packets



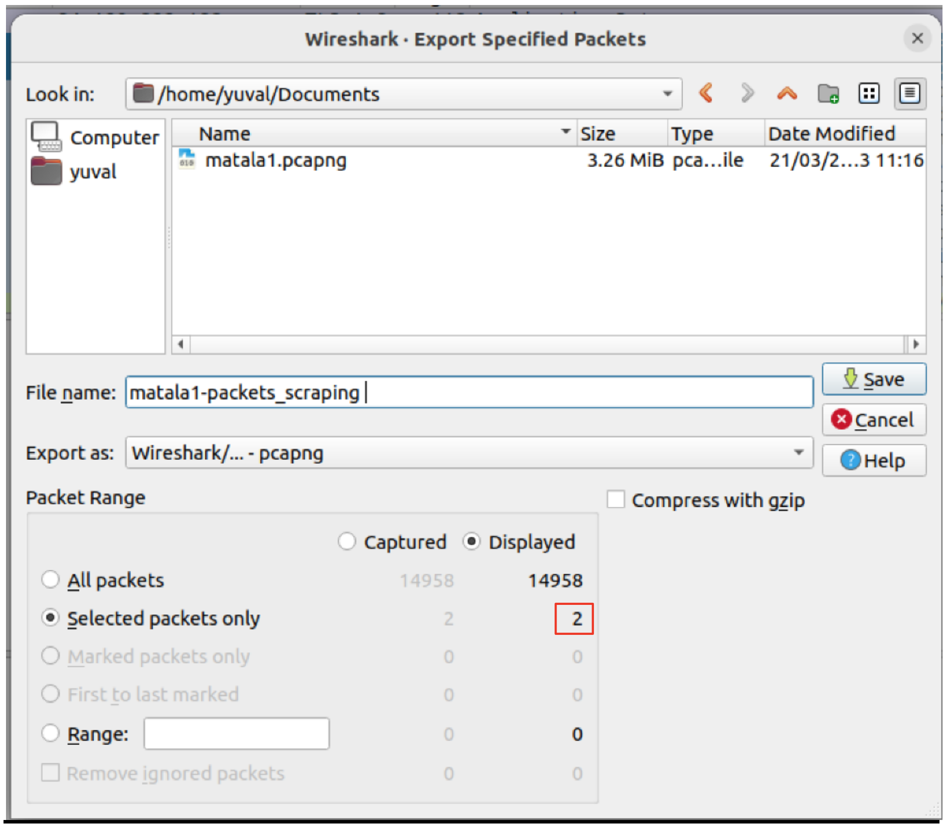
Question number 3:

We saved the packets in pcaping format.
.pcapng is one of the appropriate formats for saving a recording of a wireshark capture.

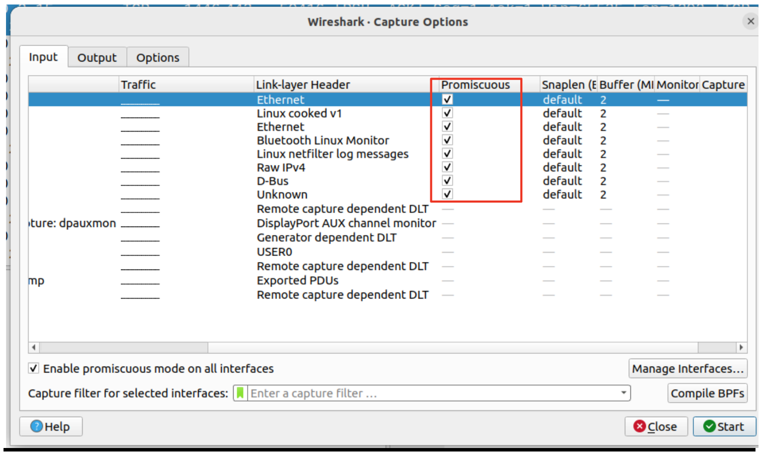


Question number 4:

As one can clearly see here, we selected 2 packets by using the shift button and selecting the packets we wanted. we can the export the specified packets that we chose.



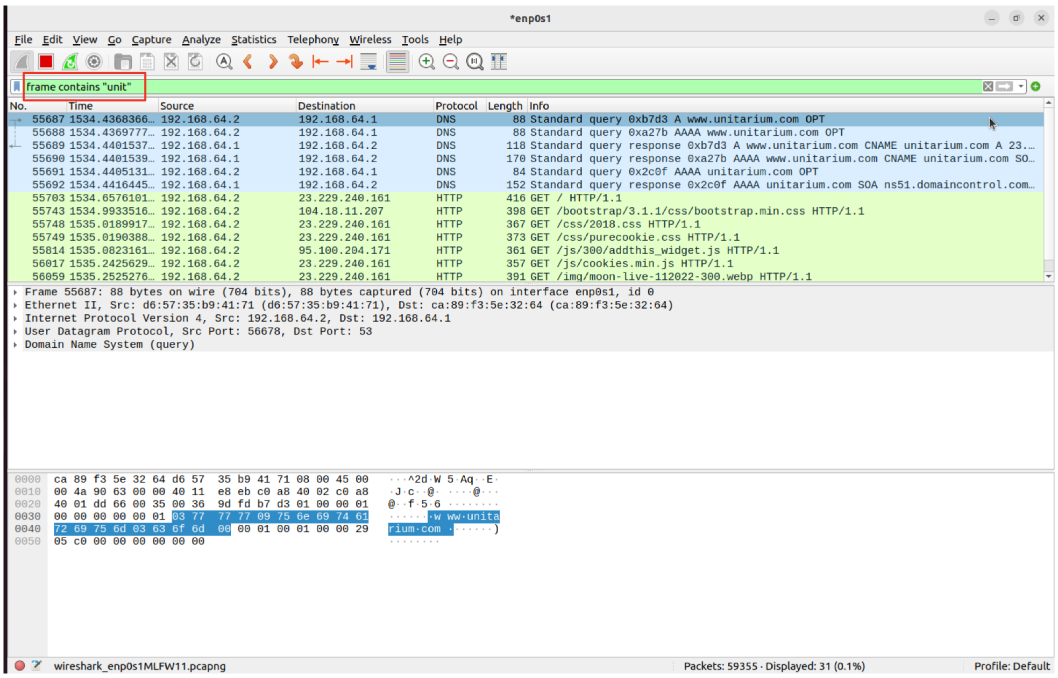
Question number 5:



As one can see we have located the promiscuous mode in the appropriate settings and checkmarked them.

a) This mode lets all network data packets be accessed and viewed by all network adapters with this mode enabled.
b) This mode is used and implemented by a program that captures all network traffic visible on all configured network adapters on a system. Unlike the regular mode, in which the sniffer filters from and to the computer its running on, in promiscuous mode, all traffic will be passed.
c) this can be used in the domains of network security, monitoring and administration

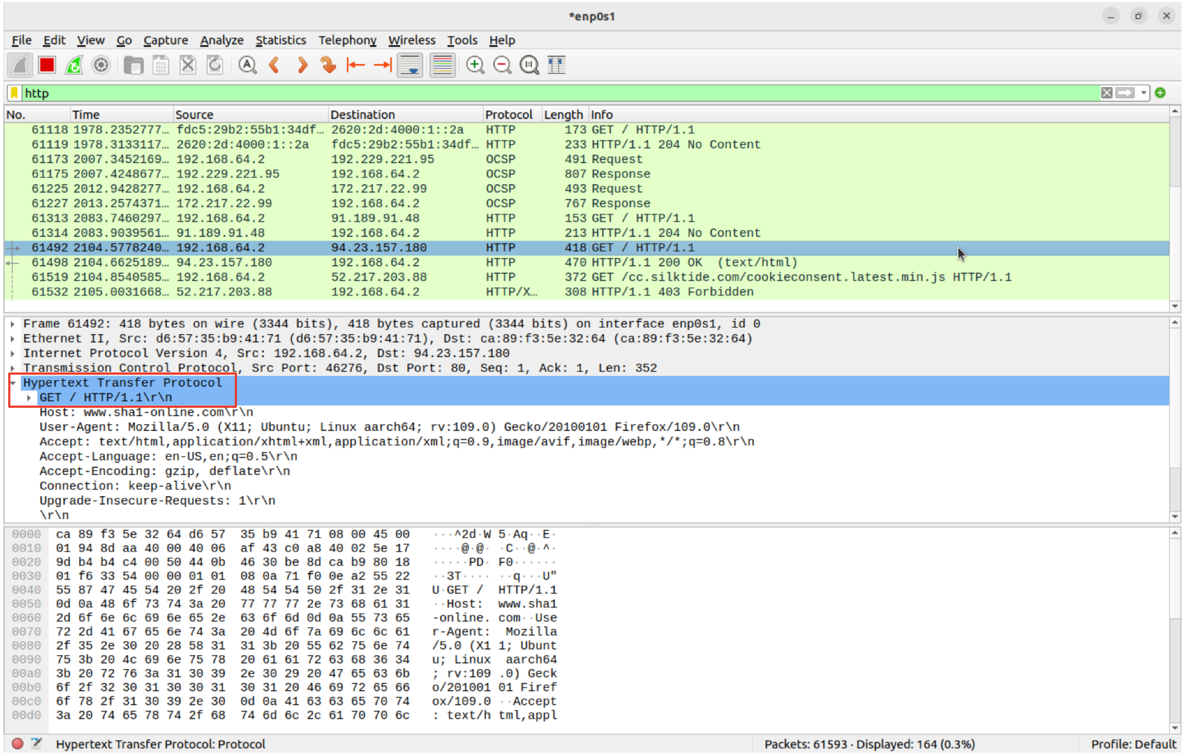
Question number 6:



The wireshark filter, filters for frames containing the string "unit", thus as a result because we entered the website stated in the question that clearly includes the string "unit" in the URL, we receive filtered results containing that website.

HTTP:

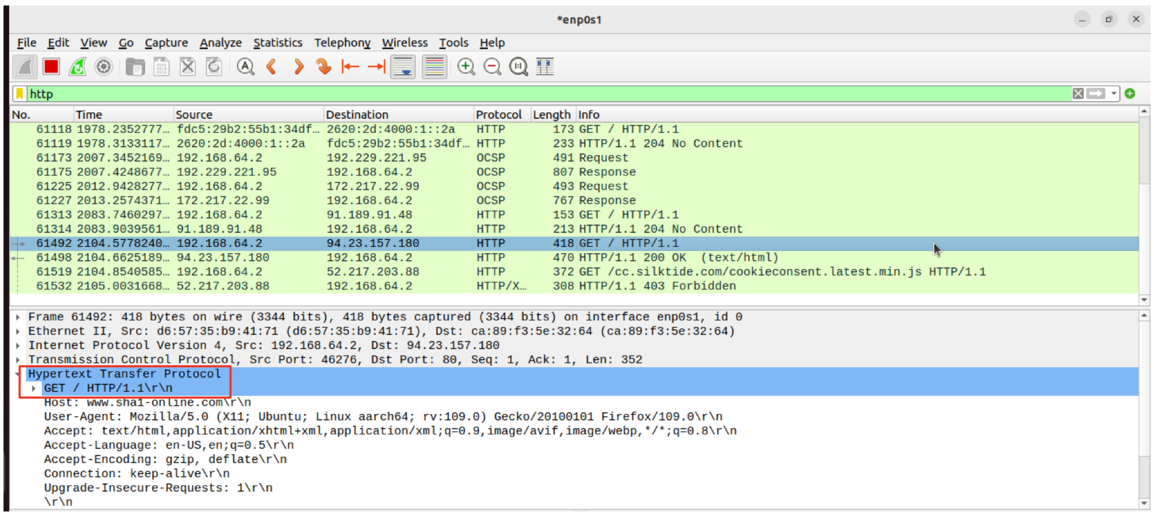
<u>שאלה מספר 7:</u>



we can identify the queries by the GET method.
In the screenshot we can deduce the request from my local IP and the ones from sha1 (des-94.23.157.180).
if we click on the packets we can clearly see the area of the hypertext and expand the contents and see the name of the host. Furthermore one can see in the screenshot that before a request there is a right pointing triangular arrow , the answer to the request is with a left pointed triangular arrow.

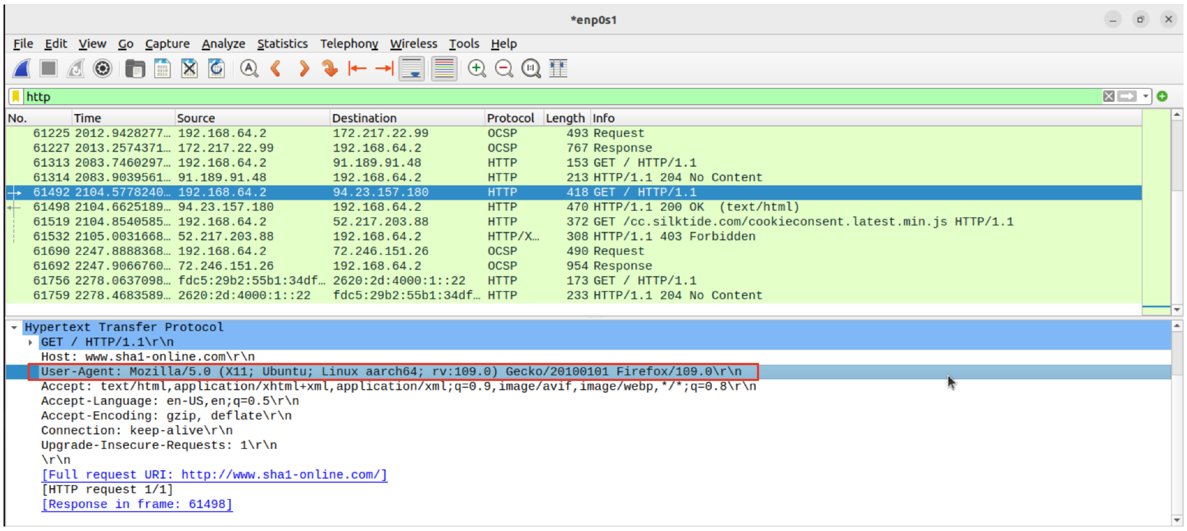
<u>Question number 8:</u>

<u>A:</u>



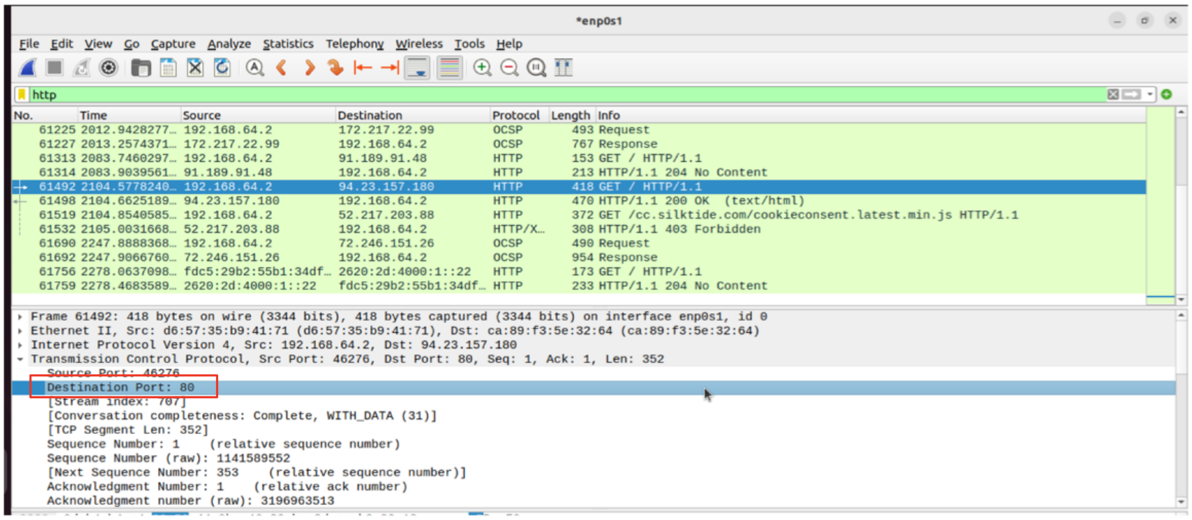
the HTTP version is stated after the GET request and is version 1.1

B:



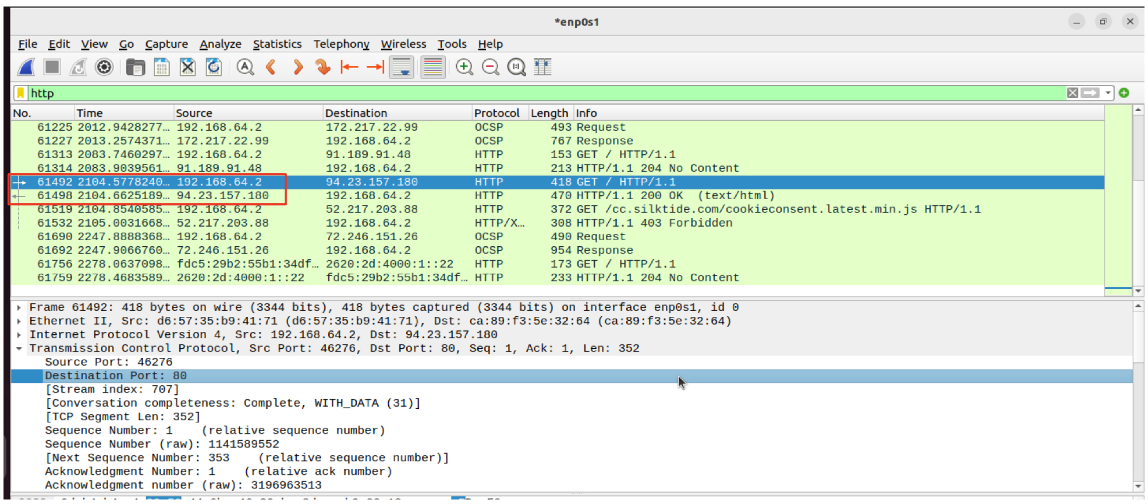
by the stated user agent we can see the request was made by ubuntu linux through firefox

C:



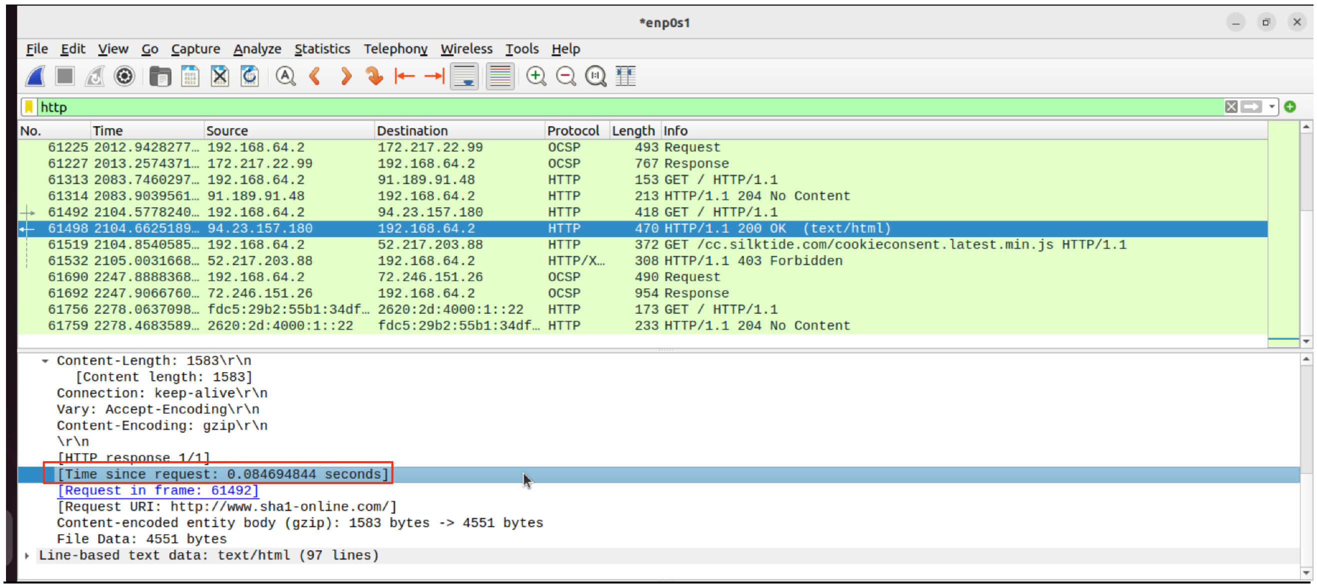
The destination port is port 80. this is the standard port for HTTP

D:



The request response package can be seen under the original request.
Furthermore, we can identify the response package by the left arrow.

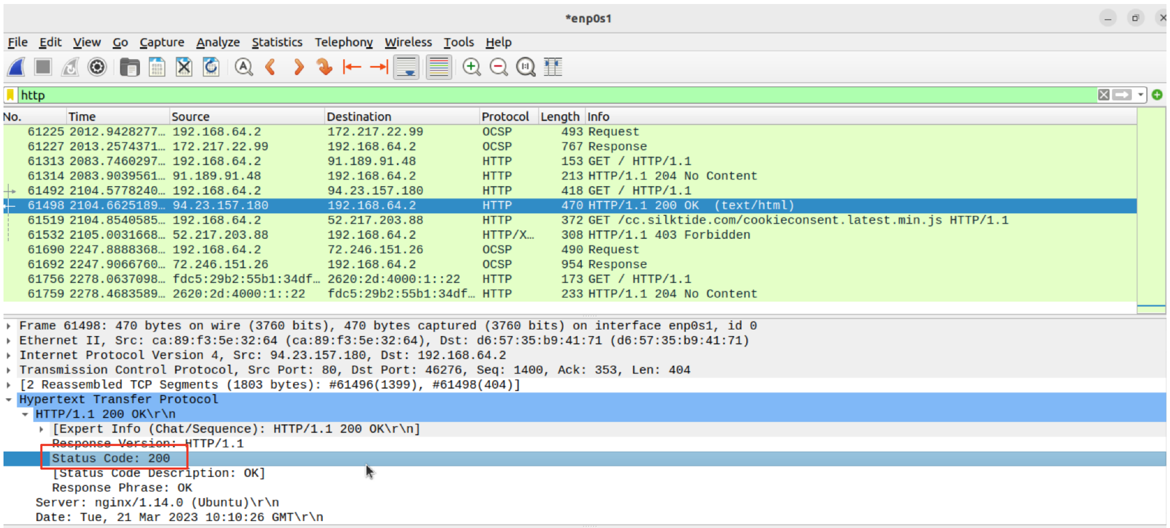
E:



0.84694844 seconds.

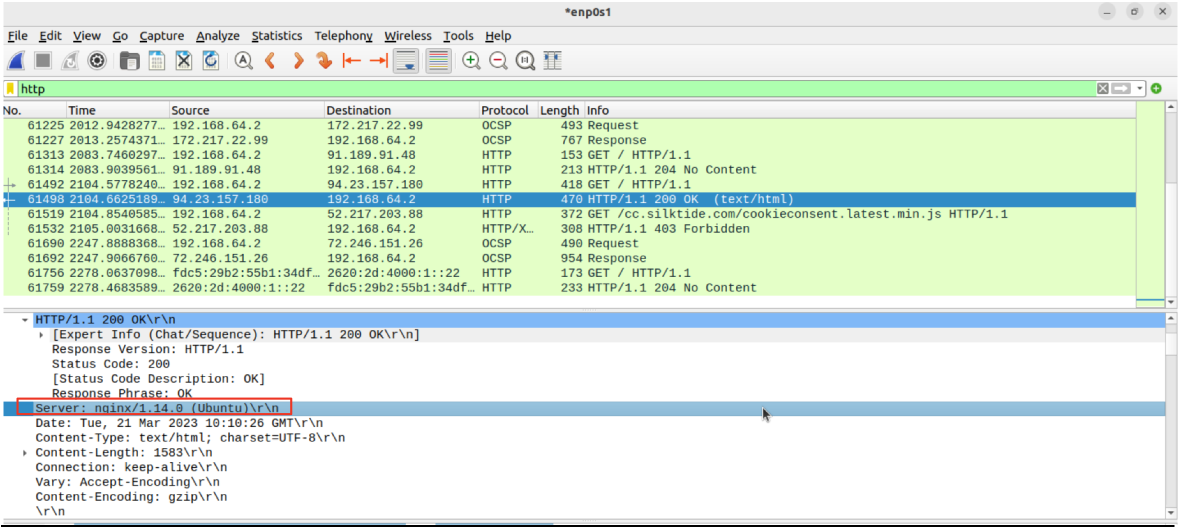
QUESTION NUMBER 9:

A:



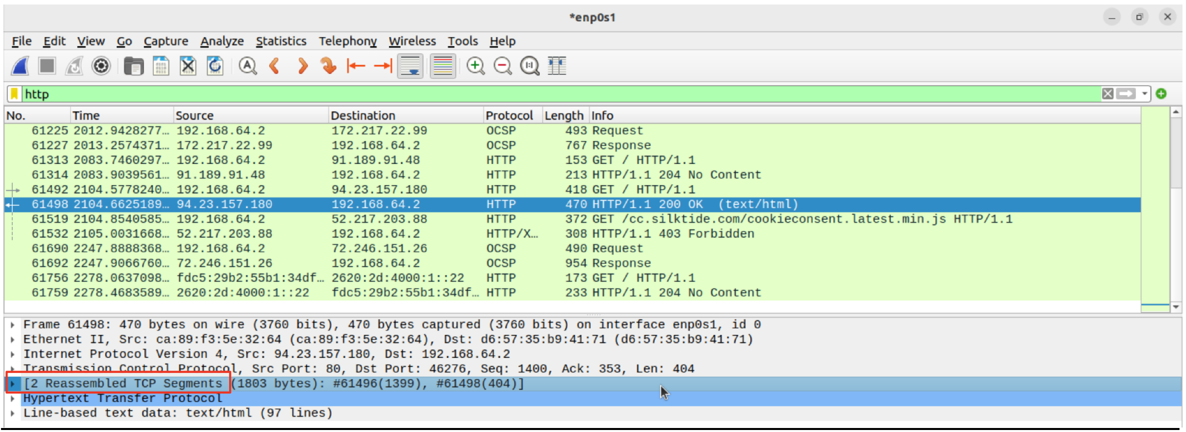
status code = 200.

B:



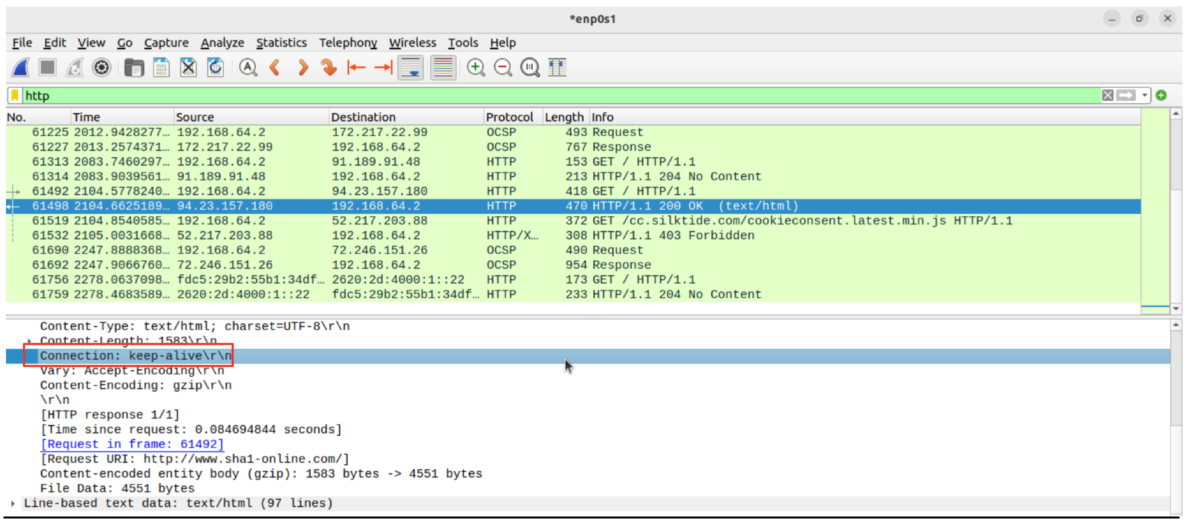
the server: nginx/ 1.14.8 (ubuntu)

C:



2 packets.

D:

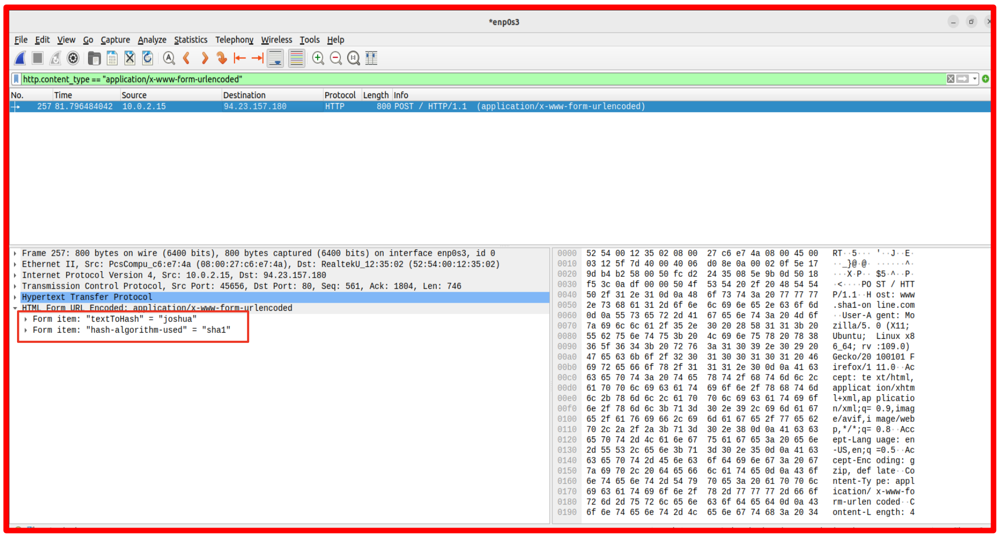


keep-alive connection means keep the connection “alive” and open. This means we don't need to connect again and again everytime we connect with the website.

QUESTION NUMBER 10:

1) The computation was computed on the server side. we come to the conclusion as the protocol was OCSP , this basically means a certificate indicating the success of the operation on the server side.

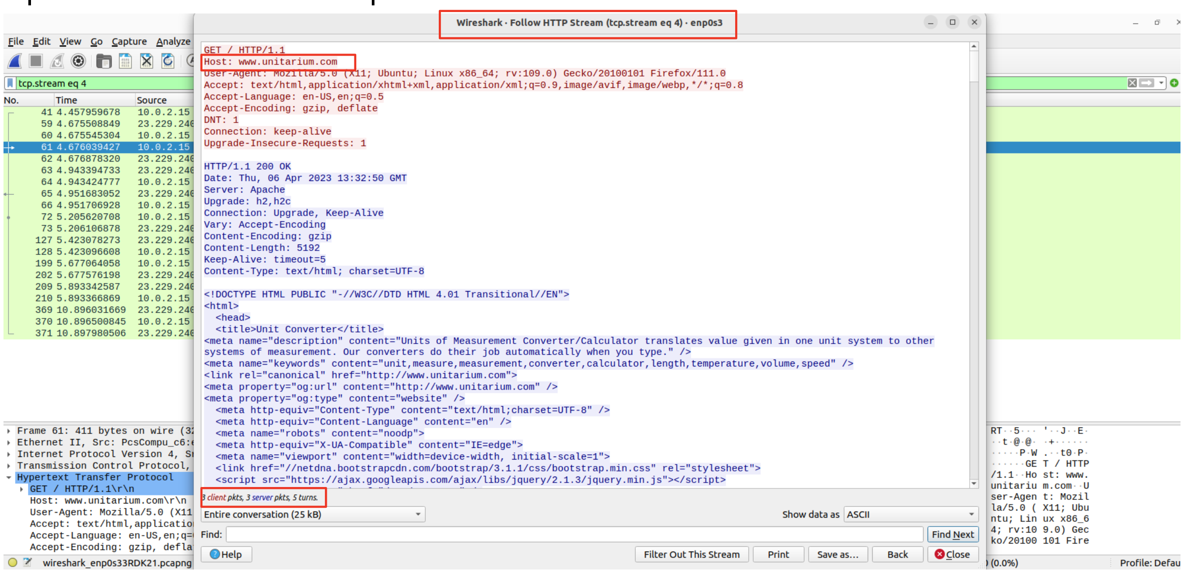
2) The sent parameters are “Joshua”, as parameter to hash, and “sha1”, as the hashing method.



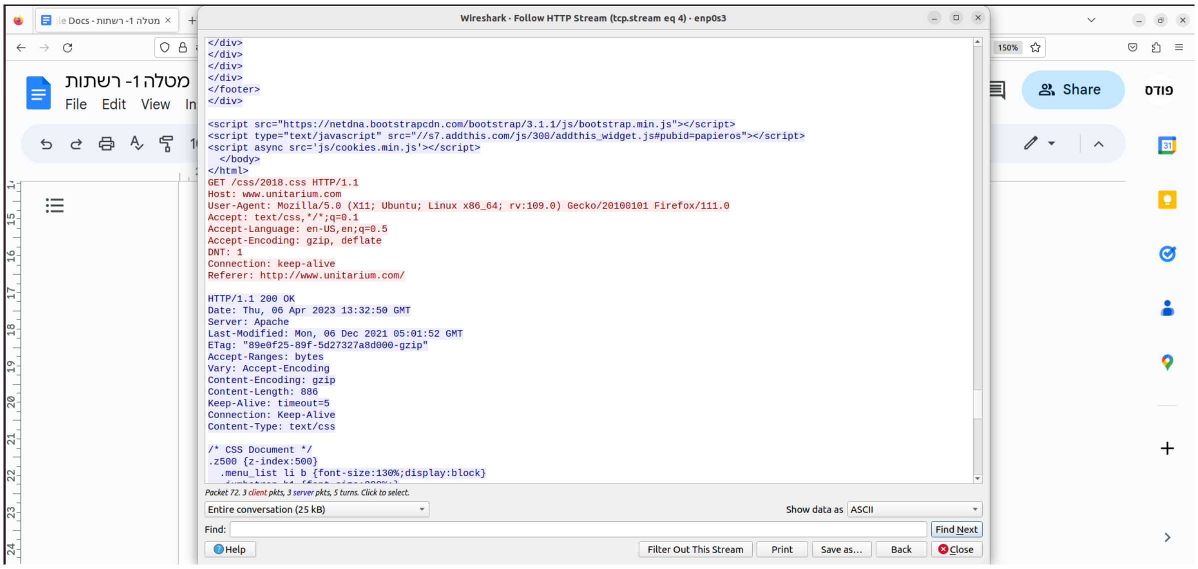
3) A reason this computation might have been done on the server instead of the browser is the computational power needed to complete the sha1 algorithm is intense and needs a large amount of computing power.
4) One danger could be that “sniffers” could intercept the data being sent back. if the answer was done in the browser no sniffers would intercept the answer.

Question number 11:

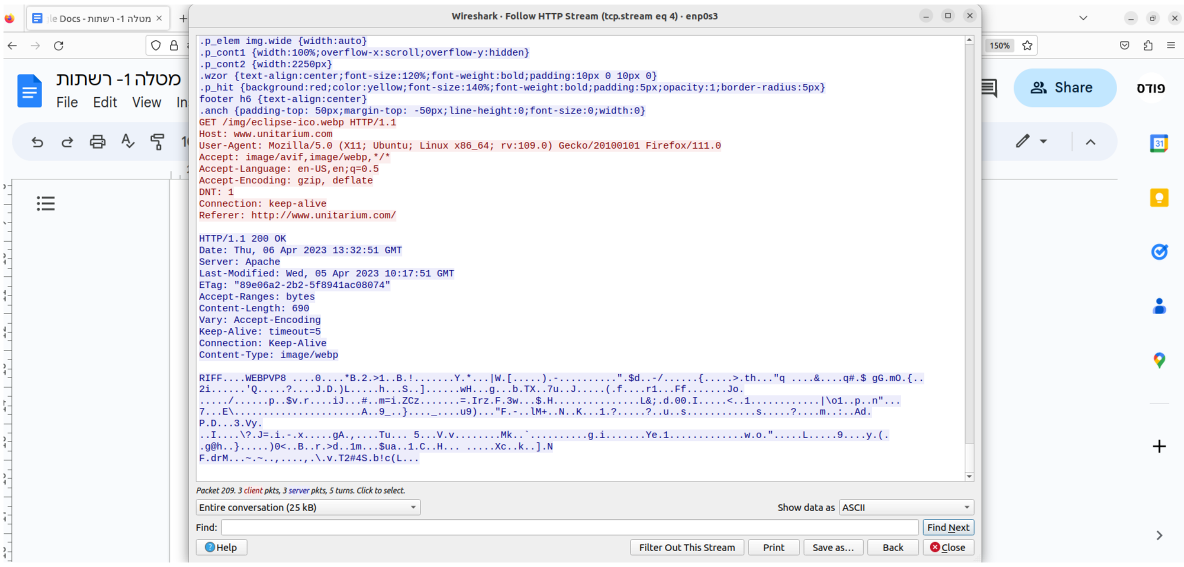
3 packets from client and 3 packets from server 3 were returned from the server.



the first packet contained the html document as shown above
the second package contained the css (the file that decorates the html file)



the third package contained a image in ico.webp format



Question number 12:

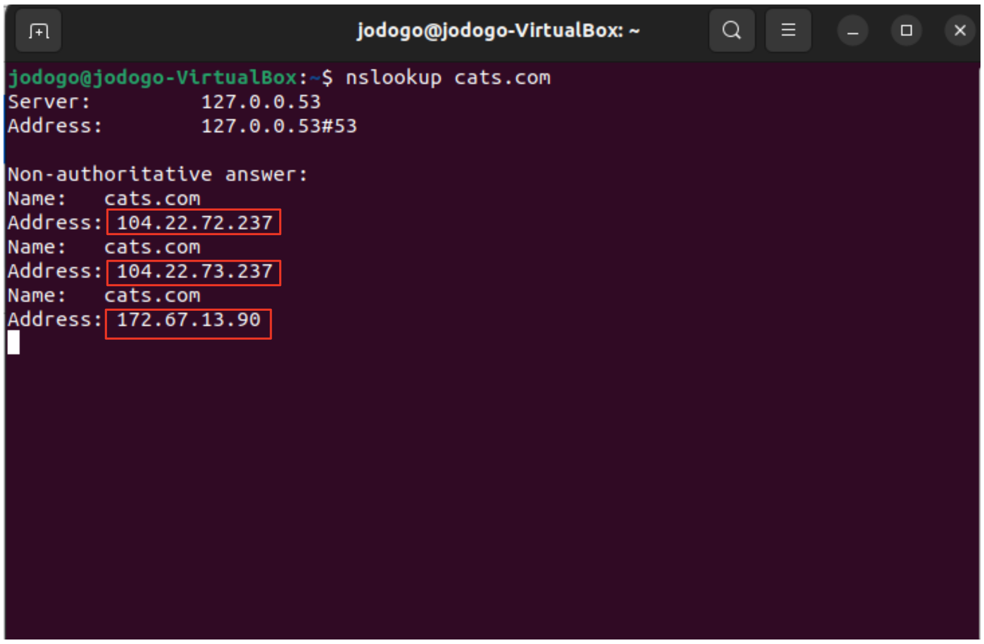
We entered amazon.com
then there were multiple DNS queries trying to find amazons ip address (trying in local catch and then searching local and wider networks). once found there was an http get request with a return value of moved permanently. then there were other dns queries sent until we finally got the page back and loaded. (it looked like amazon was registering a key for our new entry for our session).

DNS

Question number 13:

A:

no this is not an authoritative server. the server is 127.00.00.00 this is our local server as we are working on a virtual machine this answer came up.



B:

the three ip address's for the domain are:

104.22.72.235

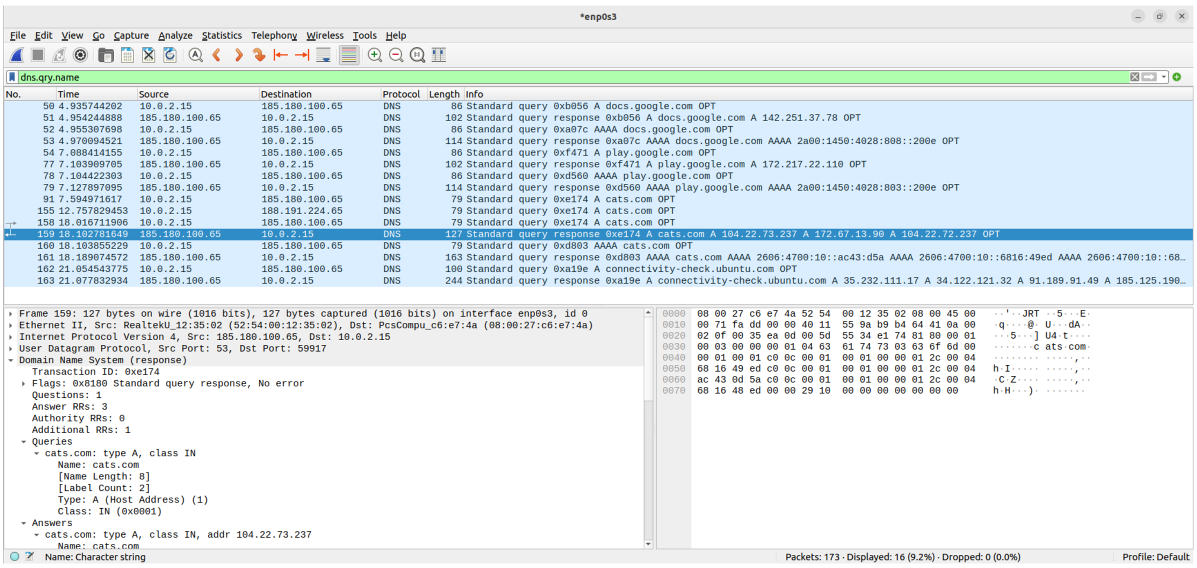
104.22.73.237

172.67.13.90

Question number 14:

A:

4 queries

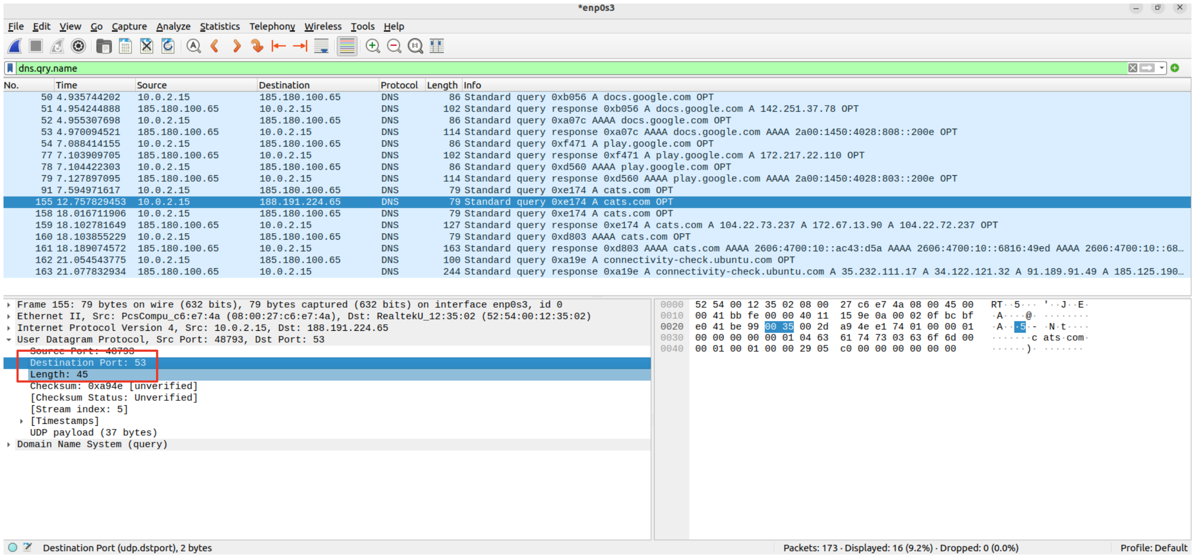


B:

the first 3 are get the response of each of the ip addresses, the last one get a general response for all 3 ip address. The queries are A query, meaning a question or a request for information expressed in a formal manner.

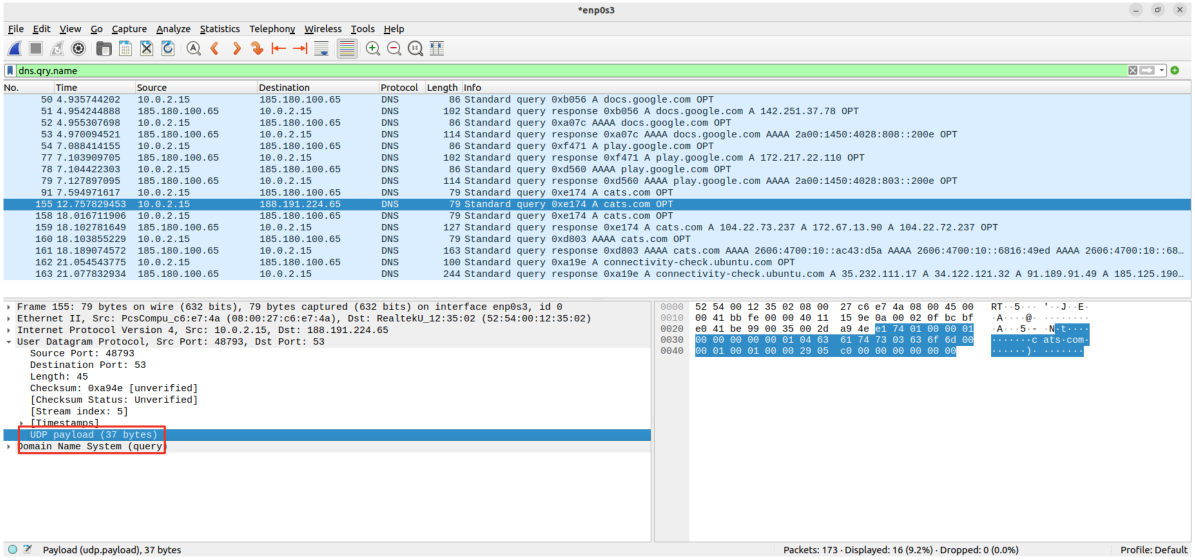
<u>C:</u>

port 53:



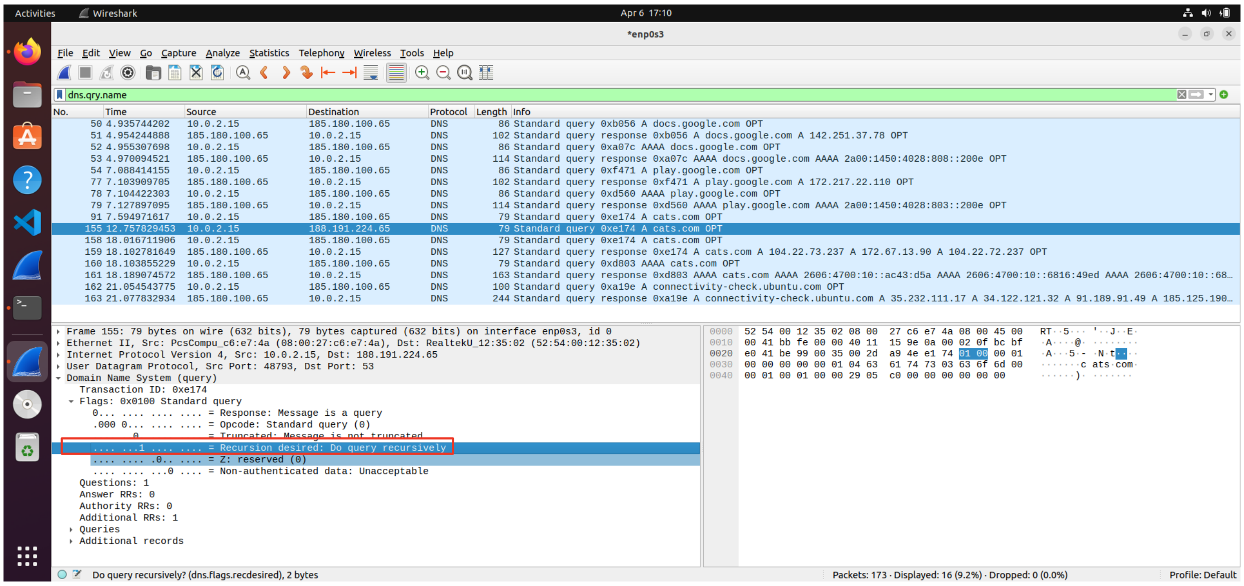
<u>D:</u>

UDP



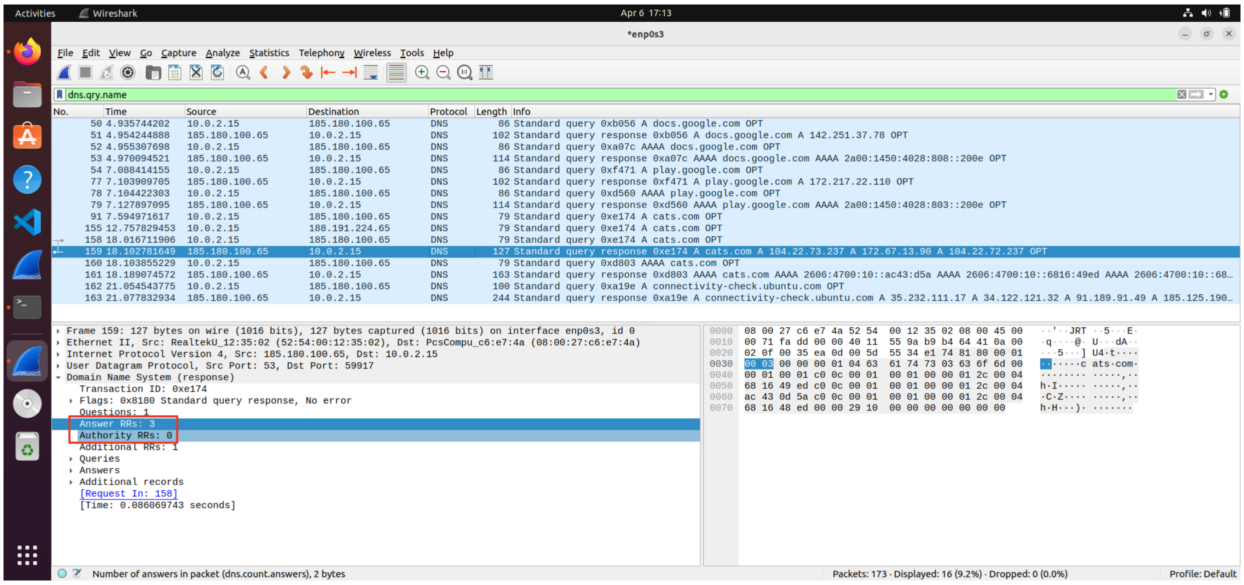
E:

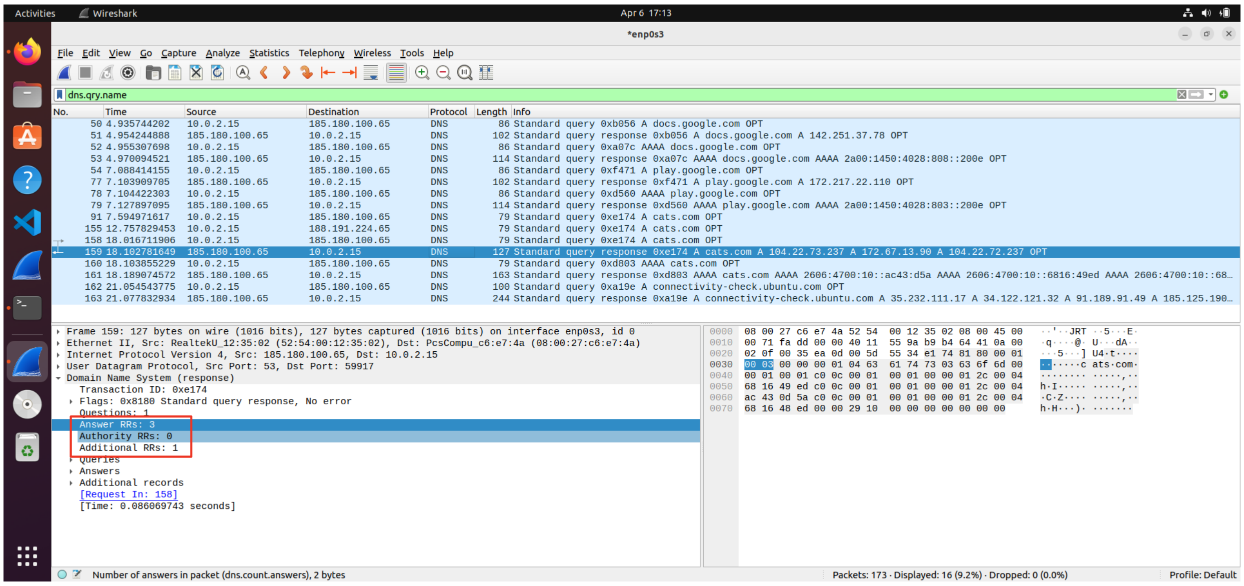
the do query recursively flag is on so it is done recursively: this can be seen by the true value of the bit in the screensot.



F:

the first three queries received a response containing 3 answers , as did the last query response:

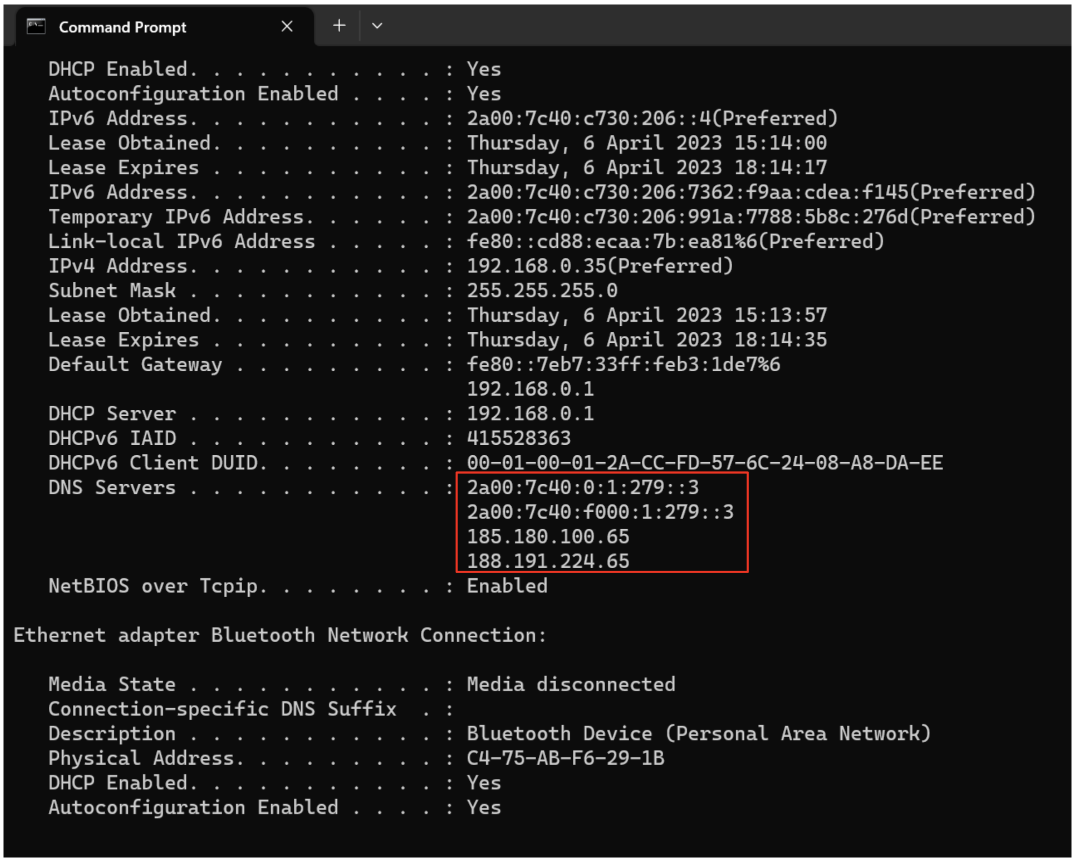




G:

A query is used to resolve a nameserver which uses IPv4. AAAA query is used to resolve a nameserver which uses IPv6.

Question number 15:



```
Command Prompt

   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2a00:7c40:c730:206::4(Preferred)
   Lease Obtained. . . . . . . . . . : Thursday, 6 April 2023 15:14:00
   Lease Expires . . . . . . . . . . : Thursday, 6 April 2023 18:14:17
   IPv6 Address. . . . . . . . . . . : 2a00:7c40:c730:206:7362:f9aa:cdea:f145(Preferred)
   Temporary IPv6 Address. . . . . . : 2a00:7c40:c730:206:991a:7788:5b8c:276d(Preferred)
   Link-local IPv6 Address . . . . . : fe80::cd88:ecaa:7b:ea81%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.0.35(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, 6 April 2023 15:13:57
   Lease Expires . . . . . . . . . . : Thursday, 6 April 2023 18:14:35
   Default Gateway . . . . . . . . . : fe80::7eb7:33ff:feb3:1de7%6
                                       192.168.0.1
   DHCP Server . . . . . . . . . . . : 192.168.0.1
   DHCPv6 IAID . . . . . . . . . . . : 415528363
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2A-CC-FD-57-6C-24-08-A8-DA-EE
   DNS Servers . . . . . . . . . . . : 2a00:7c40:0:1:279::3
                                       2a00:7c40:f000:1:279::3
                                       185.180.100.65
                                       188.191.224.65
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : C4-75-AB-F6-29-1B
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

DNS server:

2a00:7c40:0:1:279::3

2a00: 7c40:f000:1:279::3

185.180.100.65

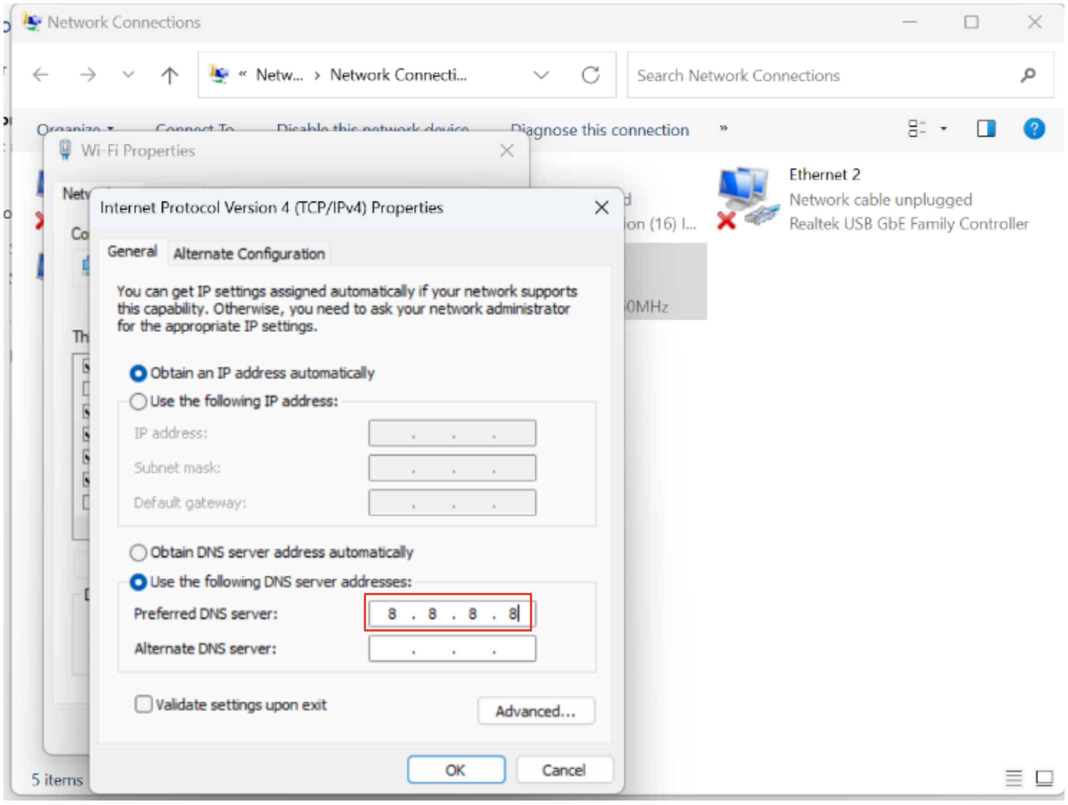
188.191.224.65

Question number 16:

likewise in the above picture we can see:
IPv4: 192.168.0.35
IPv6: 2a00:7c40:c730:206::4

Question number 17:

properties: changing to 8.8.8.8



new DNS server:

```
                                        192.168.0.1
   DHCP Server . . . . . . . . . . . : 192.168.0.1
   DHCPv6 IAID . . . . . . . . . . . : 415528363
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2A-CC-FD-57-6C-24-08-A8-DA-EE
   DNS Servers . . . . . . . . . . . : 2a00:7c40:0:1:279::3
                                        2a00:7c40:f000:1:279::3
                                        8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

2a00: 7c40:0:1:279::3
2a00:7c40:f000:1:279::3
8.8.8.8

Question number 18:

Without loss of generality take for an example some query to some web application say amazon. further without loss of generality let the query be “ mens black socks”. this query is a string and is quite small compared to the product listings you would receive from the response, as the response would return image of products , descriptions ect… there for the response is usually much heavier compared to the request.

Question number 19:

A:

common websites we vist on the device

```
Command Prompt

Microsoft Windows [Version 10.0.22621.1413]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Joshua>ipconfig/displaydns

Windows IP Configuration

    dns.google
    ----------------------------------------
    Record Name . . . . . : dns.google
    Record Type . . . . . : 28
    Time To Live  . . . . : 630
    Data Length . . . . . : 16
    Section . . . . . . . : Answer
    AAAA Record . . . . . : 2001:4860:4860::8888


    Record Name . . . . . : dns.google
    Record Type . . . . . : 28
    Time To Live  . . . . : 630
    Data Length . . . . . : 16
    Section . . . . . . . : Answer
    AAAA Record . . . . . : 2001:4860:4860::8844


    dns.google
    ----------------------------------------
    Record Name . . . . . : dns.google
    Record Type . . . . . : 1
    Time To Live  . . . . : 607
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
```

B:

```
C:\Users\Joshua>ipconfig/flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\Joshua>
```

No DNS in catch memmory as we flushed before the memmory

```
Command Prompt

Microsoft Windows [Version 10.0.22621.1413]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Joshua>ipconfig/displaydns

Windows IP Configuration


C:\Users\Joshua>
```

C:
after pinging cats.com the memory of catch is now not empty and contains cats.com

```
Command Prompt


C:\Users\Joshua>ping cats.com

Pinging cats.com [2606:4700:10::6816:48ed] with 32 bytes of data:
Reply from 2606:4700:10::6816:48ed: time=92ms
Reply from 2606:4700:10::6816:48ed: time=89ms
Reply from 2606:4700:10::6816:48ed: time=94ms
Reply from 2606:4700:10::6816:48ed: time=83ms

Ping statistics for 2606:4700:10::6816:48ed:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 83ms, Maximum = 94ms, Average = 89ms

C:\Users\Joshua>ipconfig/displaydns

Windows IP Configuration

    cats.com
    ----------------------------------------
    Record Name . . . . . : cats.com
    Record Type . . . . . : 28
    Time To Live  . . . . : 285
    Data Length . . . . . : 16
    Section . . . . . . . : Answer
    AAAA Record . . . . . : 2606:4700:10::6816:48ed


    Record Name . . . . . : cats.com
    Record Type . . . . . : 28
    Time To Live  . . . . : 285
```
we can see the ip address of cats.com is and also the time to live is 285

QUESTION NUMBER 20:

The stages of surfing the internet at a website http is the following:

us as humans remember URLs and it would be really hard to remember IP addresses, therefore we type in a search engine connected to the intent of a url of a website. say for example wix.com.

a series of queries is then looked up in DNS tables to find the value of the ip address in regards to the URL. the search is first looked up in our catch and if its not there it recursively looks up higher levels until it eventually gets to a ISP . if at that stage a ip address is not found we will receive an error. if an ip address is found a GET http request is sent and we try to receive a response. if no response was given then this could mean the server is down or is having other issues. if we receive a response a connection is opened on a keepalive basis, we make a GET http request for the html file, css files and java script (as in regards to html3) where html can be described as the skeleton of the website, css the decoration of the skeleton amd javascript the functionality of the skeleton. although a html.index page that is returned may have multiple files on it such as links and images the response usually “fires back all of these as well” some websites use lazy loading inorder to render and reduce loading time as this can significantly hurt conversion rates and SEO as google does not reward high bounce rates and consumers have been introduced to a standard of up to 3 seconds loading. The html , css and java script are then returned and run in the web application browser . the connection is closed if there is no interaction for a predetermined amount of time.